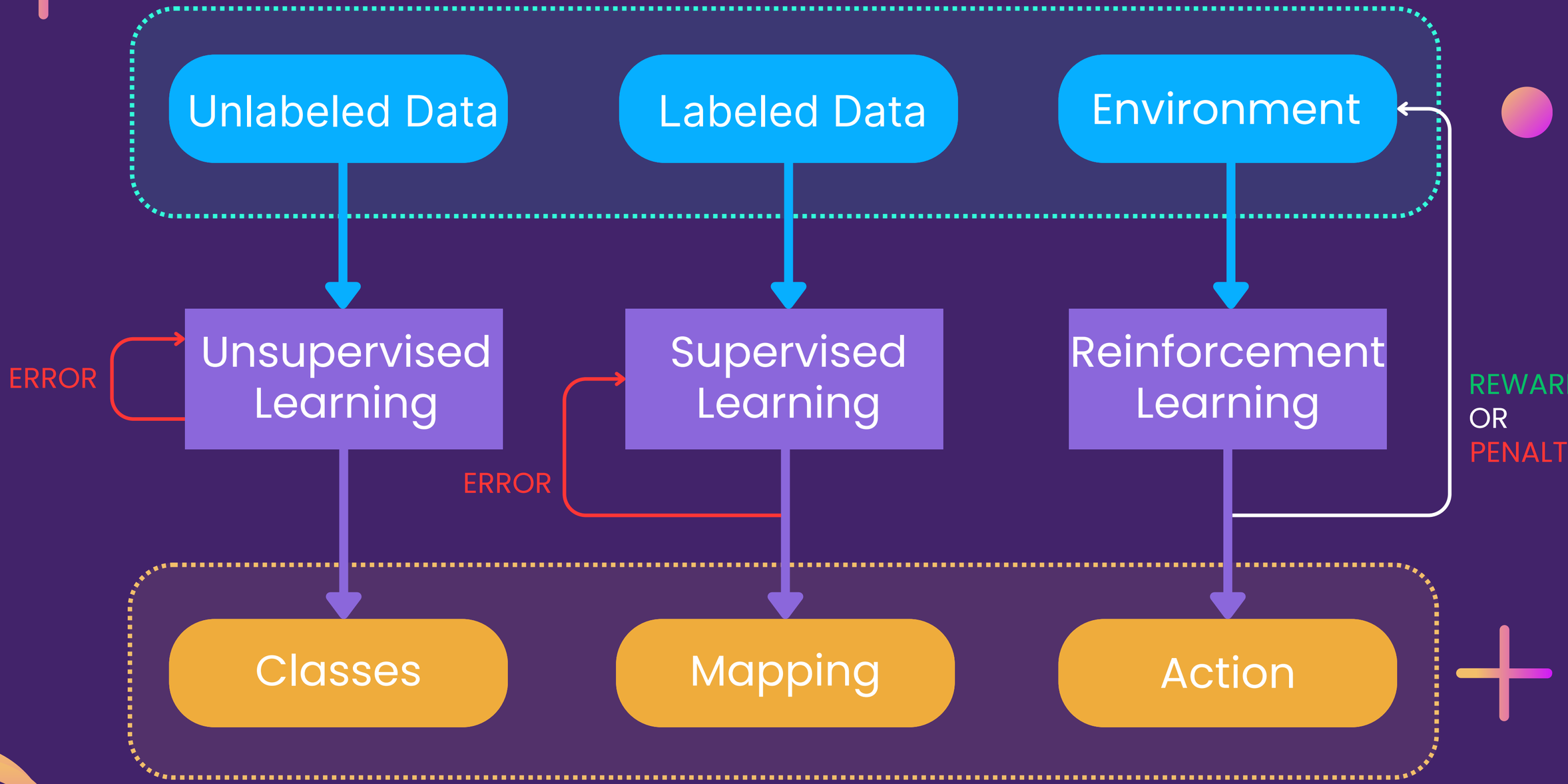

ALGORITHMIC BIAS
exploring bias in machine learning

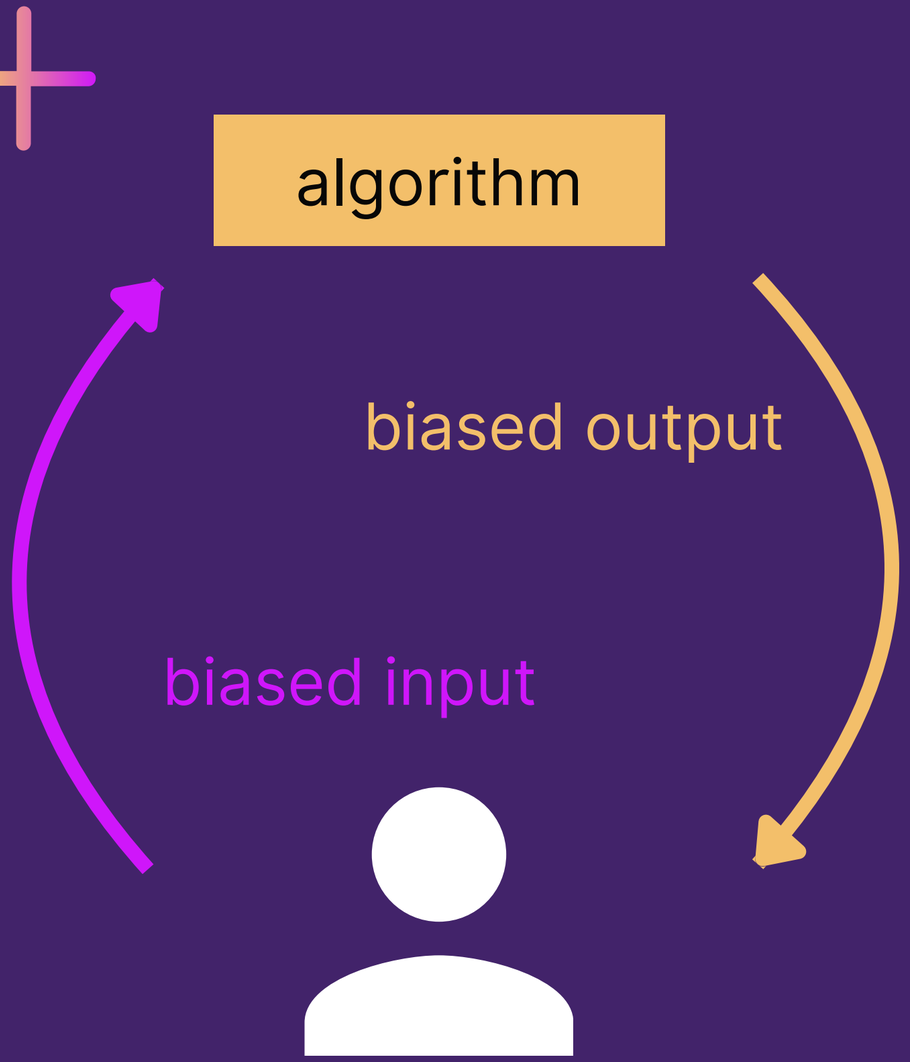
How do Machines Learn?



What is Algorithmic Bias?

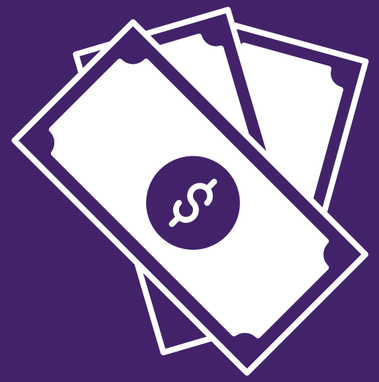
A deviation in output, performance, or impact, relative to a standard.

— Professor Sina Fazelpour, "Algorithmic bias: Senses, sources, solutions"





Discriminatory healthcare practices



Unequal housing, loans, credit, and job applications



Disproportionate surveillance and arrests of specific demographics

World



Sampling bias and lack of representation in datasets



Patterns of bias and discrimination incorporated into data distributions

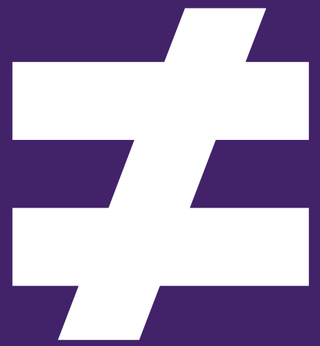


Incomplete & outdated data sources

Data



Exclusionary design, model building, and testing



Biased deployment and monitoring



(Un)conscious bias in feature selection

Design



Exacerbates quality-of-life inequality and demographic treatment



Hazardous & discriminatory repurposing of biased systems

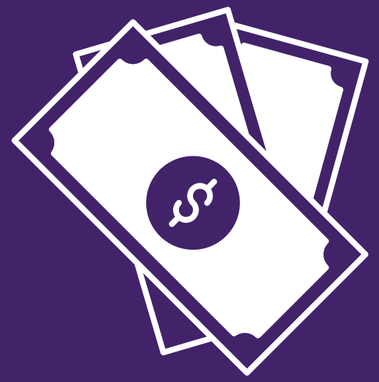


Reinforcement of societal stereotypes

Application



Discriminatory healthcare practices



Unequal housing, loans, credit, and job applications

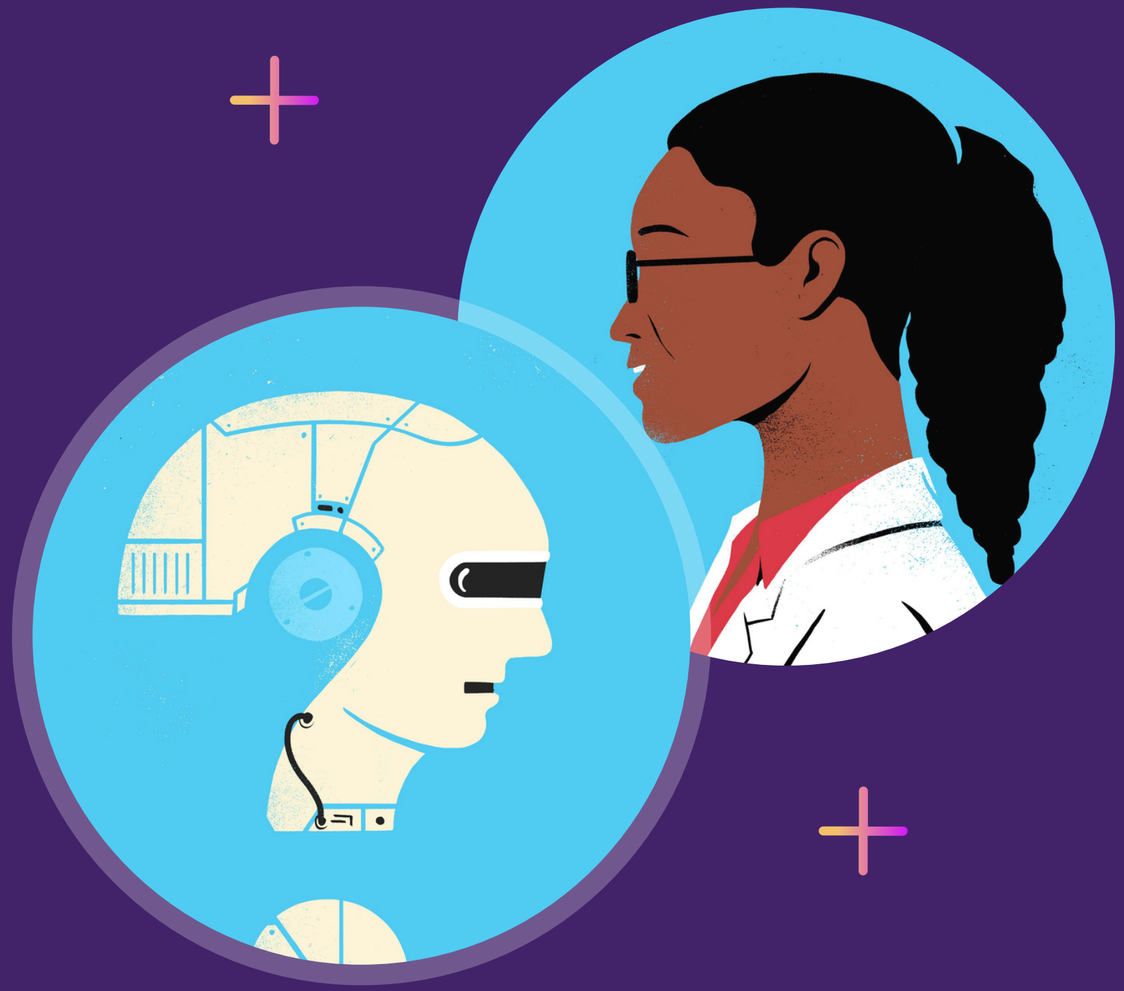


Disproportionate surveillance and arrests of specific demographics

World

Artificial Intelligence as a Mirror

AI can amplify societal biases just as mirrors can magnify imperfections



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